



University of Piraeus

School of Information and Communication Technologies

Department of Digital Systems

Level: Master's Programme of Studies

Wireless and Mobile Network Security

Case 2

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Abstract

Transition from monitoring to exploitation through a targeted Deauthentication attack in a controlled home lab environment. The attack was carried out using Kali Linux and the TP-Link TL-WN722N adapter, forcing a client device to disconnect from the wireless network.

Abstract

Transition from monitoring to exploitation via a targeted deauthentication attack in a controlled environment. Initial identification of the target router's physical address (BSSID) and transmission channel utilizing the airodump-ng tool. Subsequent injection of forged management frames to spoof the Access Point's identity and forcefully disconnect the target device. Final confirmation of the 802.11 standard vulnerabilities and the lack of authentication mechanisms in management frames, highlighting the network's susceptibility to Denial of Service (DoS) attacks.

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Introduction

This assignment is a continuation of Use Case 1, where through passive monitoring and analysis of wireless traffic at the 802.11 frame level, the Management Frames mechanism was studied. Use Case 2 moves to active exploitation of this mechanism.

Privacy Note: For security and personal data protection reasons, certain sensitive identifying details (portions of MAC addresses and internal IP addresses) have been intentionally obscured in the screenshots.

Practical part of the assignment

In this stage of the exercise, the transition from monitoring (sniffing) to intervention (packet injection) is carried out. The Deauthentication attack exploits a fundamental weakness of the 802.11 protocol: Management frames, including Deauthentication, are not authenticated.

Analysis of the scan results led to the successful identification of the target personal network (SSID 'JK') and confirmation that the target client device was connected to it. Analysis of the terminal recording captures the Deauthentication attack in full progress. Specifically, aireplay-ng was used with the -0 option (Deauthentication), targeting the client's MAC address on the target access point's channel.

The success of the Deauthentication attack is fully confirmed by analysing the monitoring data via the airodump-ng tool.

```
(kali@kali)-[~]
$ sudo airodump-ng -c 11 --bssid 30:DE:4B: wlan0
```



```
CH 11 ][ Elapsed: 4 mins ][ 2026-04-04 15:10 ][ WPA handshake: 30:DE:4B:
BSSID          PWR RXQ Beacons  #Data, #/s CH  MB  ENC CIPHER AUTH ESSID
30:DE:4B:      -23  0    1603    411   0  11  130  WPA2 CCMP PSK  JK
BSSID          STATION          PWR   Rate    Lost  Frames  Notes  Probes
30:DE:4B:      78:8B:2A:      -48   1e- 1e    0     173
30:DE:4B:      54:48:E6:      -44   6e- 6e    2      64
30:DE:4B:      84:7C:9B:      -46   1e- 1e   337   2398  EAPOL  JK
```

Conclusion

In the second and final stage of the exercise, the transition from passive monitoring to active intervention was carried out through a targeted Deauthentication attack.

Channel monitoring during the attack via airodump-ng confirmed the complete collapse of the target client's connection to the access point, with the client reappearing as 'not associated'.

This assignment demonstrates in practice that the Deauthentication attack is not simply a DoS method for causing a temporary disruption. It constitutes a fundamental stepping stone for more complex attacks, such as the capture of the WPA2 4-way handshake — which is the object of the next use case.

Bibliography

- Wikipedia contributors. (2026, April 4). Wi-Fi deauthentication attack. Retrieved from https://en.wikipedia.org/wiki/Wi-Fi_deauthentication_attack
- Zieniūtė, U., & Zieniūtė, U. (2026, January 6). What is a deauthentication attack, and how does it work? Retrieved from <https://nordvpn.com/blog/deauthentication-attack/?srsltid=AfmBOorMmQBhE-2qXtcE9vW-rjKnuid9BwTAfTB2jjtd8R3A5yE779su>
- Iliadis, F., & Iliadis, F. (2024, March 18). Deauthentication Attacks | Baeldung on Computer Science. Retrieved from <https://www.baeldung.com/cs/deauthentication-attacks>

- GeeksforGeeks. (2024, September 19). WiFi deauthentication attack against 802.11 protocol. Retrieved from <https://www.geeksforgeeks.org/linux-unix/wi-fi-deauthentication-attack-against-802-11-protocol/>
- David Bombal. (2024, July 28). *Cracking WiFi WPA2 Handshakes (And does it work with WPA3?)* [Video file]. Retrieved from <https://www.youtube.com/watch?v=X49lIPHcurE>
- User. (n.d.). Which part of the eapol packets contain WPA password Hash? Retrieved from <https://security.stackexchange.com/questions/41467/which-part-of-the-eapol-packets-contain-wpa-password-hash>
- Molenaar, R., & Molenaar, R. (2026, March 27). WPA and WPA2 4-Way Handshake. Retrieved from <https://networklessons.com/wireless/wpa-and-wpa2-4-way-handshake>